

Jack F. Cooper

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https://jackcooper-dotcom.github.io/engineering-portfolio/

EDUCATION

University of Maryland

Bachelor of Science in Mechanical Engineering

GPA: 3.91/4.00 | Dean's List: Every Semester

Relevant Coursework: Dynamics, Physics III, Calculus III, Differential Equations, Introduction to CAD

College Park, MD

Expected May 2029

PROJECTS

Autonomous Rover Project

Fall 2025

- Designed and assembled an autonomous rover using Arduino-based control and integrated electromechanical components
- Manufactured chassis and integrated ultrasonic and piezo sensors into an autonomous rover, contributing to consistent successful completion of 5 of 6 competition objectives
- Nominated for the "Best Constructed Vehicle Award"

Sheet Metal Roller Design

Spring 2026

- Modeled a 30+ part sheet metal roller assembly in SolidWorks, applying GD&T standards and accounting for press-fits and manufacturing tolerances
- Performed FEA stress analysis and motion simulations to validate structural performance before manufacturing

EXPERIENCE

Needham Department of Public Works

Needham, MA

Water Treatment Plant Assistant

June 2026-August 2026

- Conducted weekly water quality testing across 12 municipal sampling locations, measuring chlorine concentration and temperature to ensure compliance with health standards
- Maintained treatment plant equipment and grounds using industrial power equipment while adhering to OSHA and municipal safety procedures

Needham Park and Recreation

Needham, MA

Summer Camp Counselor

June 2023-August 2025

- Supervised daily activities for groups of 20+ children, ensuring safety and engagement
- Collaborated with staff to organize, schedule, and supervise 7 group activities daily

Needham House of Pizza

Needham, MA

Cashier/Delivery Driver

May 2022-May 2026

- Managed high-volume in-person and phone orders, accurately processing customer transactions and large group orders
- Delivered orders efficiently, safely, and on time, while providing excellent customer service

SKILLS & CERTIFICATIONS

- CAD: SolidWorks (CSWA), Fusion 360
- Programming: MATLAB, Arduino
- Manufacturing: 3D Printing, Laser Cutting, Basic Machining